Fiscal Year:	FY 2004	**Task Last Updated:**	FY 07/07/2004
PI Name:	Hoban-Higgins, Tana M Ph.D.		
Project Title:	Gravity and the Insect Circadian Timing System		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	95616	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fuller, Charles (University of California Davis) Wassmer, Gary (Bloomsburg University)		
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Performance Goal Text:			
	The circadian timing system (CTS) coordinated temporal aspects of physiology and behavior. Disruptions in circadian timing not only adversely affect an organism's ability to respond to environmental challenges, but also decrease performance and contribute to psychological disorders in humans. Circadian rhythm problems are identified as a risk factor for human performance failure in NASA's Critical Path Research Plan. Exposure to altered force environments affects circadian function. In an experiment carried out as part of the NASA/Mir program in Fundamental Biology, we found that the period of the insect circadian clock was changed during exposure to altered force environments. The period of the clock was shorter in microgravity than in 2G produced via centrifugation. The data also suggested that adaptation of the CTS to microgravity may follow a long timeline. In recent experiments conducted in our lab, to rodent models with neurovestibular deficits, did not have the usual circadian responses to exposure to 2G via centrifugation.		

Task Description:	This raises the question of whether or not the circadian effects seen under altered ambient force environments are part of a response to input from gravitational sensory receptors. This research program will examine the response of the insect CTS to long-term exposure to altered ambient force environments. We will measure effects on period and determine the phase-shifting response to light pulses in normal insects and in insects with reduced gravitational sensing. The data from these studies will significantly add to our understanding of the role of gravity on this fundamental physiological system. Further, these experiments on this simple biological system will contribute to our understanding of issues relating to biomedical problems of spaceflight. Throughout this period, we have continued to work with the engineers who are adapting the hardware that would be used to perform this experiment on the International Space Station. We have continued to test the hardware as needed. We have performed a ground-based experiment in support of this research project. In order to determine the effect of reduced gravitational input on the response of the insect CTS to exposure to a hyperdynamic field, we recorded activity rhythms from a group of Blue death-feigning beetles (Cryptoglossa verrocussus) under 1g and under 2g produced via centrifugation. We then removed the bristles of the PGR system from the legs of these insects and again recorded activity rhythms under both g-levels. These data again confirm an effect of g upon circadian period and suggest that reducing the gravitational input may also affect the circadian timing system. Biological clocks are ubiquitous in living organisms. They are found in every eucaryote thus far examined, and, more recently, in procaryotes. Although the first biological rhythms experiment was performed in 1729, it is only in the last 50 years that interest in the study of biological rhythms has grown rapidly. The circadian timing system (CTS) is responsible for the temporal coordination of an organism’s physiology and behavior both internally, i.e. within the body, and with the external environment, i.e. the 24-hour day. As such, the circadian timing system influences almost all physiological and behavioral functions. Studies examining the molecular basis for circadian rhythms have revealed remarkable similarities in species ranging from Drosophila to mammals. Humans had been thought to be unaffected by external light-dark cycles. However, we now know that sufficient bright light will suppress human melatonin secretion, entrain and phase shift human circadian rhythms, as well as affect the period of the human clock. This, coupled with the discovery of various chronobiologic disorders in humans has increased interest in circadian rhythm research. The CTS has been implicated in such phenomena as jet-lag, performance decrements, the problems associated with shift work, delayed sleep phase insomnia and some forms of depression. Through experiments conducted in the microgravity environment on the Mir Space Station and at 2G produced via centrifugation, we were able to show that the period of the insect circadian pacemaker is affected by the ambient force environment. The period of the activity rhythm was longer in the 2g produced during centrifugation than in the microgravity spaceflight environment. The data also suggested that exposure to microgravity may affect the sensitivity of the CTS to light exposure. The goal of this research program is to identify how the gravitational environment affects CTS function. Having this knowledge would allow the design of countermeasures to gravitationally induced circadian dysfunction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Biological clocks are ubiquitous in living organisms. They are found in every eucaryote thus far examined, and, more recently, in procaryotes. Although the first biological rhythms experiment was performed in 1729, it is only in the last 50 years that interest in the study of biological rhythms has grown rapidly. The circadian timing system (CTS) is responsible for the temporal coordination of an organism's physiology and behavior both internally, i.e. within the body, and with the external environment, i.e. the 24-hour day. As such, the circadian timing system influences almost all physiological and behavioral functions. Studies examining the molecular basis for circadian rhythms have revealed remarkable similarities in species ranging from Drosophila to mammals. Humans had been thought to be unaffected by external light-dark cycles. However, we now know that sufficient bright light will suppress human melatonin secretion, entrain and phase shift human circadian rhythms, as well as affect the period of the human clock. This, coupled with the discovery of various chronobiologic disorders in humans has increased interest in circadian rhythm research. The CTS has been implicated in such phenomena as jet-lag, performance decrements, the problems associated with shift work, delayed sleep phase insomnia and some forms of depression. The goal of this research program is to identify how the gravitational environment affects CTS function. Having this knowledge would allow the design of countermeasures to gravitationally induced circadian dysfunction.
Task Progress:	Throughout this period, we have continued to work with the engineers who are adapting the hardware that would be used to perform this experiment on the International Space Station. We have continued to test the hardware as needed. We have performed a ground-based experiment in support of this research project. In order to determine the effect of reduced gravitational input on the response of the insect CTS to exposure to a hyperdynamic field, we recorded activity rhythms from a group of Blue death-feigning beetles (Cryptoglossa verrocussus) under 1g and under 2g produced via centrifugation. We then removed the bristles of the PGR system from the legs of these insects and again recorded activity rhythms under both g-levels. These data again confirm an effect of g upon circadian period and suggest that reducing the gravitational input may also affect the circadian timing system.
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